A clustering framework for conditional extremes models

Patrick O'Toole, Christian Rohrbeck, Jordan Richards

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Same (present), tense throughout if possible

Abstract

Conditional extreme value models have proven useful for analysing the joint tail behaviour of random vectors. Conditional extreme value models describe the distribution of components of a random vector conditional on at least one exceeding a suitably high threshold, and they can flexibly capture a variety of structures in the distribution tails. One drawback of these methods is that model estimates tend to be highly uncertain due to the natural scarcity of extreme data. This motivates the development of clustering methods for this class of models; pooling similar within-cluster data drastically reduces parameter estimation uncertainty.

While an extensive amount of work to estimate conditional extremes models exists in multivariate and spatial applications, the prospect of clustering for models of this type has not yet been explored. As a motivating example, we explore tail dependence of meteorological variables across multiple spatial locations and seek to identify sites which exhibit similar multivariate tail behaviour. To this end, we introduce a clustering framework for conditional extremes models which provides a novel and principled, parametric methodology for summarising multivariate extremal dependence.

In a first step, we define a dissimilarity measure for conditional extremes models based on the Jensen-Shannon divergence and common working assumptions made when fitting these models. One key advantage of our measure is that it can be applied in arbitrary dimension and, as opposed to existing methods for clustering extremal dependence, is not restricted to a bivariate setting. Clustering is then performed by applying the k-medoids algorithm to our novel dissimilarity matrix, which collects the dissimilarity between all pairs of spatial sites.

A detailed simulation study shows our technique to be superior to the leading competitor in the bivariate case across a range of possible dependence structures and uniquely provides a tool for clustering in the multivariate extremal dependence setting. We also outline a methodology for selecting the number of clusters to use in a given application. Finally, we apply our clustering framework to Irish meteorological data.

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1 Introduction

Where to describe Vignotto method? in introduction?

2 Motivating examples

2.1 Irish meteorological data

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2.2 US urban air pollution

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Will have to mention that other paper that does bivaraite clustering as well in background, right?

3 Methods

3.1 Peaks-over-threshold

3.2 Extremal dependence

Copy mostly from TFR, will have to include NI data

In order to characterise the dependence between extremes, we must introduce the concepts of asymptotic dependence and independence, as described in **?**. The coefficient of dependence, $\chi \in [0, 1]$, measures the degree of extremal dependence between two random variables X_1 and X_2 with corresponding CDFs F_1 and F_2, under the class of asymptotically dependent distributions. It is defined as $\chi = \lim_{u \to 1} \chi(u)$, where

See Huser paper for inspiration

$$\chi(u) = 2 - \frac{\log\left(\mathbb{P}(F_1(X_1) < u, F_2(X_2) < u)\right)}{\log\left(\mathbb{P}(F_1(X_1) < u)\right)}.$$

Coppied straight from TFR, may have to cut down

If $\chi = 0$, then X_1 and X_2 are said to be asymptotically independent, while for $\chi > 0$, they are asymptotically dependent, with greater magnitude as χ increases. Thus, χ gives the limit probability of X_1 being extreme given that X_2 is extreme, and we expect the joint probability of observing extremal events in both variables to be greater where χ is higher [**?**]. Furthermore, χ is symmetric in X_1 and X_2. Two variables can be asymptotically dependent even if they are not dependent in the classical sense. Another coefficient, $\bar{\chi} \in [-1, 1]$, measures the tail dependence between two variables where $\chi = 0$, in the case of asymptotically independent variables. This coefficient is similarly defined as $\bar{\chi} = \lim_{u \to 1} \bar{\chi}(u)$, where

expand CDF?

$$\bar{\chi}(u) = \lim_{u \to 1} \frac{2 \log \mathbb{P}(F_1(X_1) > u)}{\log \mathbb{P}(F_1(X_1) > u, F_2(X_2) > u)} - 1.$$

For asymptotic dependence, $\bar{\chi} = 1$. Under asymptotic independence, $\bar{\chi} = 0$, with $\bar{\chi}$ increasing in magnitude as the variables become more positively (for $\bar{\chi} > 0$) or negatively (for $\bar{\chi} < 0$) dependent in their extremes [**?**].

Together, $(\chi, \bar{\chi})$ provide a useful summary of extremal dependence for any pair of random variables. However, they do not fully describe the dependence structure of a multivariate distribution. For example, probabilities of jointly extreme events for two or more variables cannot be calculated from these coefficients. For this, we require a model for extremal dependence. Various approaches to this problem have been proposed, including max-stable processes, Pareto processes, Gaussian processes and copulas. However, these models are limited in their ability to model dependence and their computational feasibility, and are often conceptually difficult. A framework which provides a more flexible approach to modelling extremal dependence is the CE model of **?**.

3.3 Conditional extremes

3.4 Divergence metric

skew-geometric JS divergence is unbounded, need to include Thiagarajan version which is!

- **?** shows that inference can be made about the extremal dependence of a random vector $\boldsymbol{Y}_{-i} = \boldsymbol{Y} \backslash Y_i$ on a variable Y_i by conditioning on the exceedance of a high threshold u by Y_i, and assuming that this conditional distribution is a diagonal multivariate Gaussian distribution:

$$\boldsymbol{Y}_{-i} \mid Y_i = y_i \sim N\left(\boldsymbol{\alpha}_{|i} y_i + y_i^{\boldsymbol{\beta}_i} \boldsymbol{\mu}_{|i}, y_i^{\boldsymbol{\beta}_i} \boldsymbol{\sigma}_{|\boldsymbol{i}}\right), Y_i > u. \tag{3.1}$$

- Therefore, when seeking to compare the extremal dependence structure of two locations, a sensible approach is to compare the conditional distributions given by equation 3.1 for each variable.

- One common measure of dissimilarity between two distributions P and Q is the Kullback-Leibler (KL) divergence, defined as

$$KL(P \mid\mid Q) = \int_{-\infty}^{\infty} p(y) \log\left(\frac{p(y)}{q(y)}\right) dy. \tag{3.2}$$

where $p(y)$ and $q(y)$ are the densities of P and Q, respectively **?**.

define mid mid!

- The KL divergence is always non-negative and zero if and only if $P = Q$, and for two multivariate normal distributions $P \sim N(\mu_1, \Sigma_1)$, $Q \sim N(\mu_2, \Sigma_2)$ in $\mathbb{R}^n$ has closed form solution

$$KL(P \mid\mid Q) = \frac{1}{2}\left(\log\left(\frac{\det \Sigma_2}{\det \Sigma_1}\right) - n + \operatorname{Tr}\left(\Sigma_2^{-1}\Sigma_1\right) + (\mu_2 - \mu_1)^T \Sigma_2^{-1} (\mu_2 - \mu_1)\right), \tag{3.3}$$

as shown in **?**.

- In the case where both covariance matrices are diagonal, this simplifies to

$$KL(P \mid\mid Q) = \frac{1}{2}\sum_{i=1}^{n}\left(\frac{\sigma_{1,i}^2}{\sigma_{2,i}^2} + \frac{(\mu_{1,i} - \mu_{2,i})^2}{\sigma_{2,i}^2} - 1 - \log\left(\frac{\sigma_{1,i}^2}{\sigma_{2,i}^2}\right)\right), \tag{3.4}$$

where:

  - $\mu_1 = (\mu_{1,1}, \mu_{1,2}, \ldots, \mu_{1,n})^\top$ and $\mu_2 = (\mu_{2,1}, \mu_{2,2}, \ldots, \mu_{2,n})^\top$ are the mean vectors,
  - $\Sigma_1 = \operatorname{diag}(\sigma_{1,1}^2, \sigma_{1,2}^2, \ldots, \sigma_{1,n}^2)$ and $\Sigma_2 = \operatorname{diag}(\sigma_{2,1}^2, \sigma_{2,2}^2, \ldots, \sigma_{2,n}^2)$ are diagonal covariance matrices,
  - $\sigma_{1,i}^2$ and $\sigma_{2,i}^2$ denote the i-th diagonal element of Σ_1 and Σ_2, respectively.

- However, the KL divergence is unfortunately not symmetric, unbounded and not a true metric, making it a poor choice as a dissimilarity measure for clustering.

Comment on alpha

Say how this comes from VI

- The Jensen-Shannon (JS) divergence is a simple extension of the KL divergence, defined as

$$JS(P \mid\mid Q) = \frac{1}{2} KL(P \mid\mid M) + \frac{1}{2} KL(Q \mid\mid M), \tag{3.5}$$

where $M = \frac{1}{2}(P + Q)$ is a mixture distribution [**?**].

- The JS divergence provides a symmetric measure of dissimilarity between two distributions.
- It is bounded by $[0, \log(2)]$ and is a true metric, unlike the simple sum of the forward and backward KL divergences $KL(P \mid\mid Q) + KL(Q \mid\mid P)$.
- A problem arises from the fact that the mixture distribution M of two normals is not itself normal.
- In fact, M becomes increasingly bimodal as P and Q become more dissimilar, and so the JS divergence does not have a closed form solution for multivariate normals.
- **?** overcomes this problem by using the weighted geometric mean $G_\alpha(x, y) = x^\alpha y^{1-\alpha}, \alpha \in [0, 1]$ instead of the arithmetic mean, and the fact that the weighted product of two normals is also normal (as with any other member of the exponential family).
- From this, we get the skew-geometric JS divergence, defined as

$$JS^{G_\alpha}(P \mid\mid Q) = \frac{1}{2} KL(P \mid\mid G_\alpha(P, Q)) + \frac{1}{2} KL(Q \mid\mid G_\alpha(P, Q)). \tag{3.6}$$

- The parameter α is called a skew parameter, and controls the divergence skew between P and Q [**?**].

Look into in Thiagarajan paper where he says the reverse geometric mean is used by Deasy?

- For two multivariate normals, this gives the closed form solution

$$\begin{aligned} JS^{G_\alpha}(P \mid\mid Q) &= \frac{1}{2}\left((1-\alpha)\mu_1^T \Sigma_1^{-1} \mu_1 + \alpha \mu_2^T \Sigma_2^{-1} \mu_2 - \mu_\alpha^T \Sigma_\alpha^{-1} \mu_\alpha + \log\left(\frac{|\Sigma_1|^{1-\alpha}|\Sigma_2|^\alpha}{|\Sigma_\alpha|}\right)\right), \\ \Sigma_\alpha &= \left((1-\alpha)\Sigma_1^{-1} + \alpha \Sigma_2^{-1}\right)^{-1}, \\ \mu_\alpha &= \Sigma_\alpha \left((1-\alpha)\Sigma_1^{-1}\mu_1 + \alpha \Sigma_2^{-1}\mu_2\right), \end{aligned} \tag{3.7}$$

and for diagonal covariance matrices, **?** show that this simplifies to

define for diagonal covariance

$$JS - G(P \mid\mid Q) = \frac{1}{2} \sum_{i=1}^{n} \left(\frac{(1-\alpha)\sigma_{1,i}^2 + \alpha \sigma_{2,i}^2}{\sigma_{\alpha,i}^2} + \log\left(\frac{\sigma}{-}\right) \right)$$

- Thus we have a closed form, computationally efficient and symmetric measure of dissimilarity between any two conditional distributions given by equation 3.1 for two locations, with which we can cluster locations based on their extremal dependence structure.

Reword

Comment on choice of alpha

3.5 Clustering

3.5.1 Choice of k

4 Simulation study

- Throughout this section, we evaluate the effectiveness of our clustering method using a simulation study.
- In 4.1, a simple simulation design using a Gaussian copula to generate data is described. This data has the advantage of having theoretical asymptotic values for the CE parameters, to which we can directly compare our finite simulation estimates, as a test to ensure that our clustering is working as desired, reducing uncertainty in parameter estimates.

- In 4.2, we extend this study to a more complex simulation design using a mixture of Gaussian and t-copulas, for a variety of simple and more complex dependence structures and scenarios.
- We evaluate our clustering solution using the ARI, and in 4.2.4 ascertain the improvement of parameter estimation post-clustering using a bootstrapping scheme.

4.1 Gaussian copula

Our Gaussian copula simulation design for a single "location", nomenclature chosen to match the spatial nature of our applications, is as follows:

- We generate multivariate data from a bivariate Gaussian copula, where we control the dependence structure through its correlation parameter, ρ_{Gauss}. **?** show that for a bivariate Gaussian copula, the asymptotic CE parameters are $\alpha = \rho^2_{\text{Gauss}}, \beta = 1/2$, indicating asymptotic independence.
- We sample from this copula and transform to Laplace margins and use the CE model to estimate the bivariate extremal dependence structure at this location.

- We can easily extend this design by generating multiple locations with the same and different ρ_{Gauss} values.
- The knowledge of these parameters provides our "known" clustering solution.
- For example, we could generate data for four locations, the first cluster being two locations with $\rho_{\text{Gauss}} = 0.1$, and the second having $\rho_{\text{Gauss}} = 0.9$ for its two locations.
- We can use the CE model to estimate the bivariate extremal dependence structure at each location, and then apply the JS divergence to cluster these locations based on the similarity of the estimated CE parameters.
- These simulations can be repeated many times and we can evaluate the clustering solution using the ARI, and observe the bias and variance of our simulation estimates around their theoretical values.
- To this end, we generated data for 12 locations each with 1000 observations of two variables using the above design.
- Three known clusters were designated by having ρ_{Gauss} values of 0.1, 0.5, and 0.9 respectively for four locations each.
- Conditional dependence quantiles of 0.9 and 0.99 were used to estimate the CE model parameters before and after clustering.
- The number of clusters k was ascertained using the method described in ?

reference part of methods that describes this

- Simulations were repeated 500 times, and the results are shown in figure 1.
- The estimation of k and the ARI for this simple, well-defined example were perfect across all simulations.

double check this assertion!

- Several interesting points can be made from the results.
- Pre-clustering, the CE model estimates have greater uncertainty, especially for β, where the bias is also larger.

- This is especially the case for the higher dependence quantile, where data is scarcer, although this also corresponds to less bias, with the true values often lying within the first and third quantiles of each boxplot, as we take our data from further into the tail, where convergence to the asymptotic values is better.
- Bias and variance are lower for higher ρ_{Gauss}, which appears to also be associated with better convergence to the theoretical asymptotic values.
- Post-clustering, the same patterns emerge, but the variability in the estimates is reduced, especially in the problematic cases of estimating β and for the higher conditional quantile level, as desired.

Quote any specific values? Or are plots self-evident?

Stop alpha and beta facet labels from clipping

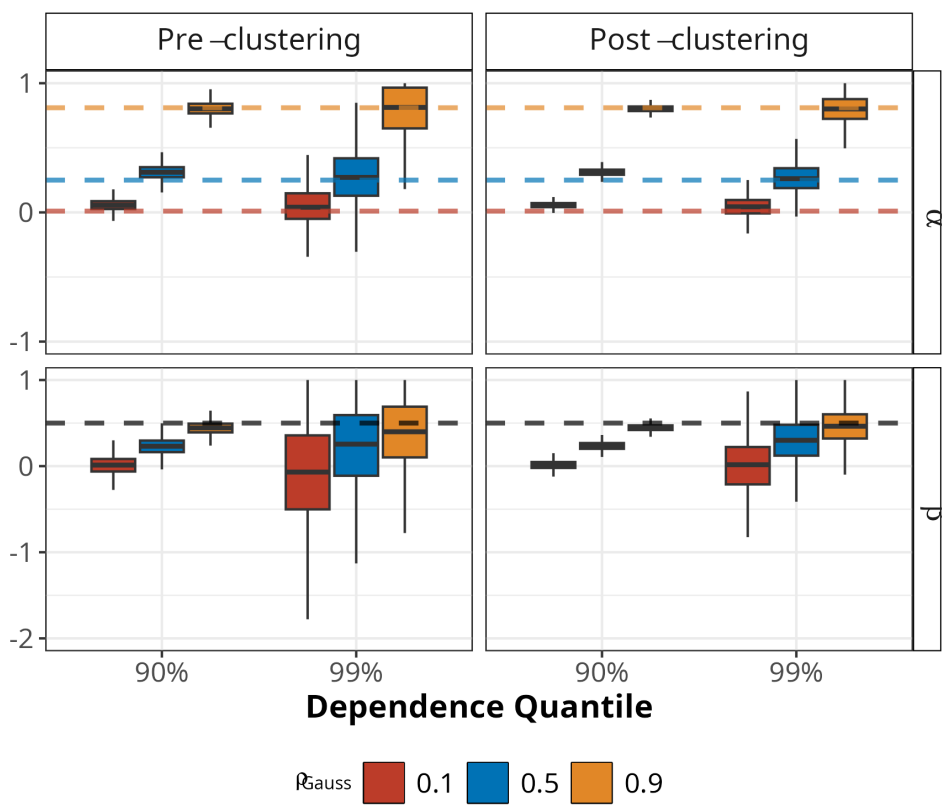

Figure 1: Boxplots of α *and* β *parameter estimates for the conditional extremes model pre- and post-clustering for a bivariate Gaussian copula simulation of 12 datasets, or "locations", each with 1000 observations of each variable, belonging equally to three known clusters. Simulations were repeated 500 times. The x-axis represents the conditional quantile used to fit the conditional extremes model, and the boxes are coloured by their Gaussian copula correlation parameter,* ρ_{Gauss}*. Dotted horizontal lines indicate the theoretical asymptotic values of* α *and* β *for a bivariate Gaussian copula, coloured by the same* ρ_{Gauss} *values for* α *and black for* β *to indicate the same value across all clusters, at* $1/2$*.*

4.2 Mixture models

- In this section, we extend our simulation design to a mixture of Gaussian and t-copulas, where we control the dependence structure through their respective correlation parameters, ρ_{Gauss} and ρ_t, with each t-copula having 3 degrees of freedom.

- The idea behind this design is that the Gaussian copula generates observations exhibiting extremal independence, whereas the t-copula induces extremal dependence, the strength of which is determined by their respective correlation parameters.
- While there are no theoretical guarantees about our estimates, we can still evaluate our clustering solution for this more difficult setting using the ARI.
- Below, we will describe the results of this simulation study in a variety of scenarios, as defined by a grid search over possible ρ_{Gauss} and ρ_t values.
- In 4.2.1, we compare our method to the leading competitor in the bivariate case, **?**, and also show how our clustering algorithm uniquely provides a tool for clustering in the multivariate extremal dependence setting in 4.2.2.
- A more realistic setting is also considered in 4.2.3, where we simulate data from 60 locations with three unequally sized clusters, and perturb the correlation parameters to make the clustering solution more challenging.
- We can also evaluate the improvement in parameter estimation post-clustering using the bootstrapping scheme described in **?**, as done in 4.2.4.

4.2.1 Comparison to competing methods

Extend to 500 simulations, only ran 200 times

- We compared our method to the leading competitor in the bivariate case, **?**, using the maximum risk function.
- As in 4.1, we generated data for 12 locations each with 1000 observations of two variables, but this time using a mixture of Gaussian and t-copulas, again with Laplace margins.
- Two known clusters were defined using different values of ρ_t, keeping ρ_{Gauss} the same for both clusters (but varying across simulations).
- The CE model was fit at the 90th dependence quantile.
- The clustering solutions were evaluated using the ARI, with results shown in figure 2 for a grid search over different ρ_{Gauss} and ρ_t values, with 500 simulations for each point.
- The proposed method is shown to be superior to the method of **?**, across a range of possible bivariate dependence structures, as defined by ρ_t.
- Both models perform better when the difference in the t-copula correlation parameters between the two clusters is larger, as the clusters are naturally more distinct when there is greater disparity in dependence.
- Both models also perform better when Gaussian and t-copula correlation are higher, as the signal of extremal dependence is stronger, with more extremal data available.

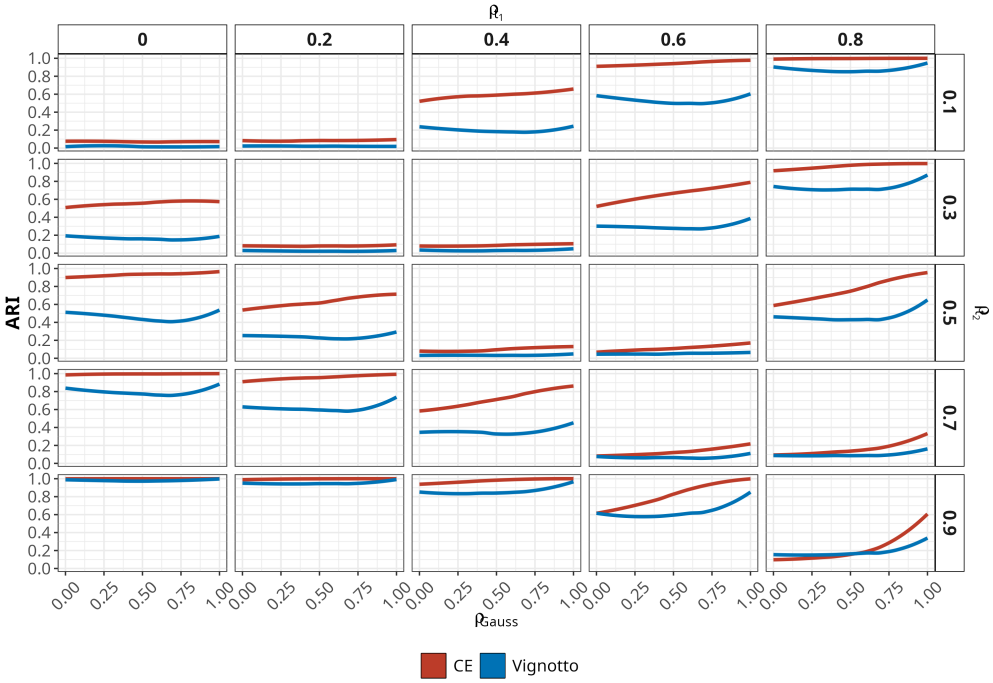

Figure 2: *Comparison of clustering methods for two variables and two equally sized clusters for simulations of 12 "locations", each with 1000 observations of each variable, from a mixture of Normal and Gaussian copulas. A grid search was performed, with the x-axis representing the Gaussian correlation parameter used for both clusters, and the facet labels showing the t-copula correlation parameters for each "known" cluster. This grid search was repeated 500 times. The lines show the median of the Adjusted Rand Index for both clustering methods, with uncertainty coming from the 90% credible interval.*

4.2.2 Extension to > 2 dimensions

- While the method in **?** is restricted to two dimensions, our method can be extended to three dimensions and beyond.
- To illustrate this, we can repeat the simulation study and grid search of 4.2.1, but now for three variables, the results of which are shown in figure 3.
- The proposed method is shown to work well for three dimensions, and the same patterns in the ARI emerge as in the bivariate case.
- The method is shown to actually perform better in three dimensions than in two.
- This comes with the caveat that in these simulations, the variables all have the same margins, and so including three variables naturally increases the amount of data available to estimate the same dependence structure as in the bivariate case, and does not introduce additional noise which might be present in real multivariate applications.

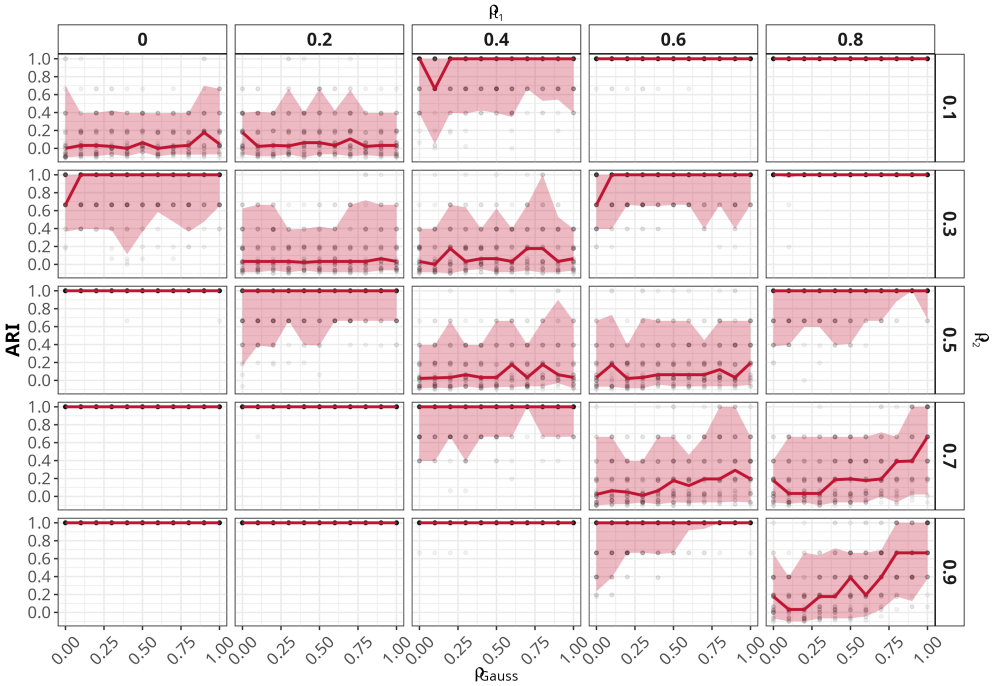

Figure 3: *Evaluation of clustering performance for three variables and two equally sized clusters for simulations of 12 "locations", each with 1000 observations for each variable, from a mixture of Normal and Gaussian copulas. A grid search was performed, with the x-axis representing the Gaussian correlation parameter used for both clusters, and the facet labels showing the t-copula correlation parameters for each "known" cluster. This grid search was repeated 500 times. The lines show the median of the Adjusted Rand Index for both clustering methods, with uncertainty coming from the 90% credible interval. Points show individual values of the ARI for a given simulation.*

4.2.3 More realistic example

- We also design a more realistic example to somewhat mimic the structure of the Irish dataset introduced in 2.1.
- We generated data from 60 locations, each with 1000 observations of two variables, using a mixture of Gaussian and t-copulas with GPD margins.
- The 60 locations had three known clusters, with 10, 20, and 30 locations in each cluster, respectively.
- A constant ρ_{Gauss} of 0.5 was used for all locations and simulations, to reduce the dimensionality of the grid search.
- A slight perturbation via a uniform sample between -0.05 and 0.05 was added to the correlation parameters of the t-copula for each location, to make the clustering solution more challenging.
- Again, clustering was done on the CE model parameters using the JS divergence, and the ARI for a grid search over different ρ_t values is shown in figure 4.
- The algorithm is again shown to perform well, particularly where the difference in the t-copula correlation parameters between the clusters is largest, as expected.

- The variability in the ARI across simulations is also quite low, indicating that the clustering solution is robust.

Check facet and x-axis labels

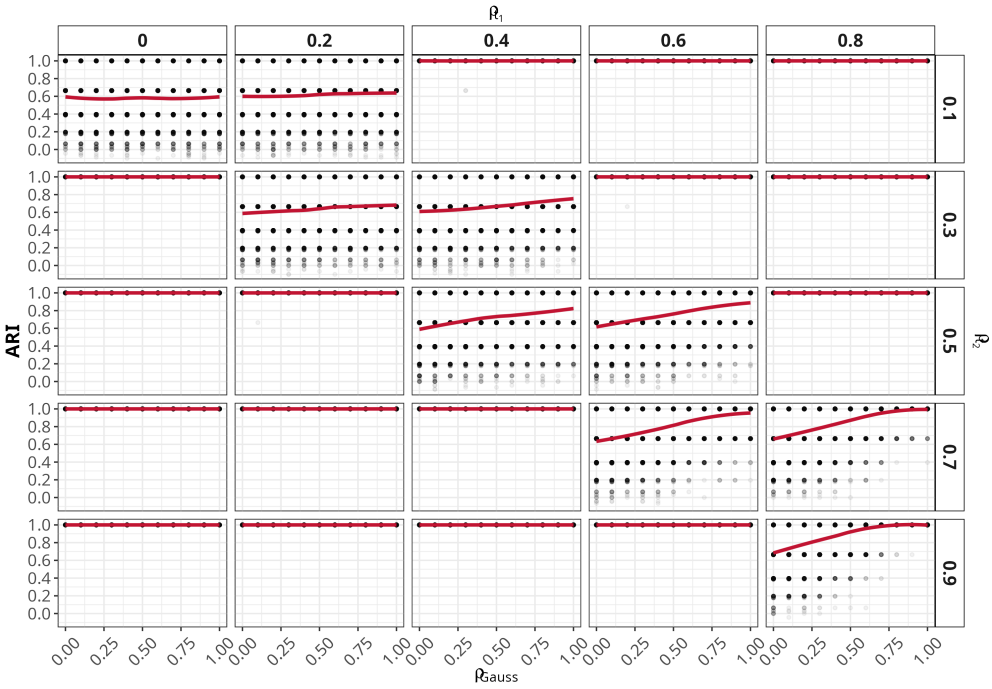

Figure 4: *Evaluation of clustering performance for two variables and three clusters for simulations from a mixture of Normal and Gaussian copulas in a more realistic clustering setting. 60 locations with 1000 observations for each variable were simulated, with 10, 20 and 30 belonging to each respective cluster. A grid search was performed, with the x-axis and facet labels showing the t-copula correlation parameters for each of the three "known" clusters, for a Gaussian copula correlation of 0.5. Perturbations were added to these parameters. This grid search was repeated 500 times. The line shows the median of the Adjusted Rand Index for both clustering methods, with uncertainty coming from the 90% credible interval. Points show individual values of the ARI for a given simulation.*

4.2.4 Parameter estimation pre- and post-clustering

- Finally, we recall that an important reason for spatial clustering is to reduce the uncertainty in parameter estimates while minimising the introduction of bias by grouping similar locations together.
- To this end, we desired to ascertain whether the estimates of our dependence parameters were less uncertain after clustering.
- We can use the bootstrapping scheme described in **?** to determine the uncertainty in our parameter estimates.
- As we are estimating two parameters, α and β, for which we do not know the true values, it may be that although our uncertainty is diminished post-clustering, the estimates may still be biased, with the parameters unidentifiable.
- There are also many possible combinations of α and β that could give the same conditional estimates, and so the estimates may be biased in this sense as well.

Do I need to describe this scheme and how it preserves the dependence structure of the data?

- To avoid this, we adopt the approach of **?** in calculating the conditional expectation of one variable given the other at the 98% marginal quantile u for the conditioning variable, given by

$$\mathbb{E}[X_1|X_2 = u] = \hat{\alpha}u + u^{\hat{\beta}}\hat{\mu}, \tag{4.1}$$

  where $(\hat{\alpha}, \hat{\beta}, \hat{\mu})$ are the estimated CE model parameters.

- We can bootstrap this conditional expectation to evaluate the reduction in uncertainty post-clustering.

Tense consistency

- We take an example for which we know the clustering algorithm performs well, namely the same simulation design as in 4.2.1, with $\rho_{\text{Gauss}} = 0.5$, and $\rho_t = 0.1$ and 0.9 for the two clusters.
- We estimate the CE model parameters pre- and post-clustering, and generate 500 bootstrap samples for the conditional expectation of 4.1.
- The results are plotted in figure 5.
- We can see that the uncertainty in the bootstrapped estimates is vastly reduced post-clustering, as desired.
- The conditional expectation is higher for the higher t-copula correlation cluster, as expected.

Anything else need to be said here? Be more specific with results?

Stop clipping for x-axis title

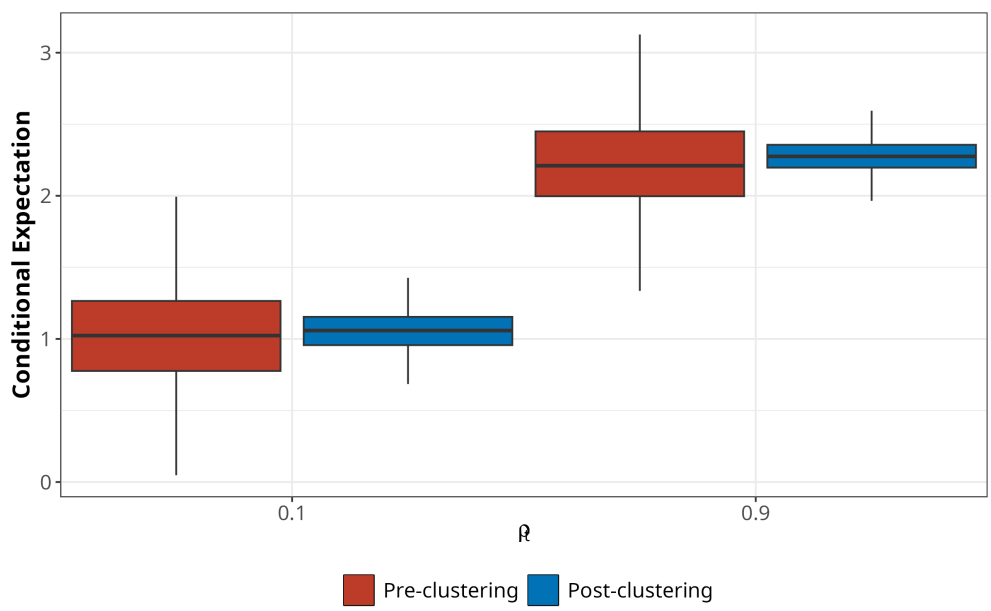

Figure 5: *Boxplots of 500 bootstrapped conditional expectations of one variable given the other is at the 98% marginal quantile u, pre- and post-clustering for a simulation of 12 "locations", each with 1000 observations for each variable, from a mixture of Normal and Gaussian copulas, with the Gaussian correlation set to 0.5. The t-copula correlation parameters for the two clusters were set to 0.1 and 0.9, as shown on the x-axis.*

5 Applications

5.1 Irish meteorological data

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5.2 US urban air pollution data

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6 Discussion

- For simulations, could have looked at LRI as well as ARI.
- Could have looked at spatially varying clustering parameters, to more closely mimic spatial applications.

Code availability

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